ICDETECTOR EXPORTS · FULL TECHNICAL REPORT

Technical report — Copia sintética de ICdetector 3.1.0

What ICdetector recorded, what CellCase checks in the data and what is worth reviewing. It does not attribute attacks.

Case	Copia sintética de ICdetector 3.1.0 — Una copia de ICdetector 3.1.0 de un teléfono que venía de 3.0.0, con datos inventados
Capture device	Teléfono sintético (sin dispositivo real) — declared by the user; not verified
Data origin	Synthetic test data
Capture time zone	Europe/Madrid — declared and confirmed assumption
Scope	1 of 1 file imported: Backup
Period	2026-10-15 – 2026-10-29 (history, phone's local time)
Generated	2026-10-02 08:05:00 UTC · CellCase 0.2.0 · episodes/6
Readers and rules	readers/2.4.0 · cellcase-rules/2 · episodes/6 · cross/3
Report language	English (en-GB); the language only changes the presentation, not the analysis

Notice

Synthetic test data: they do not describe any real phone, network or place.

1. Summary

> **Assessment for the spoofing question (the same as in the plain-language report)**
> Observations that need more data. Technical interest (review priority) is a separate criterion: the repetition of common behaviour may merit technical review without increasing suspicion.

1 priority-review incident, 2 to review and 5 context. ICdetector recorded an alarm or forensic case in 2 incidents. 14 ICdetector records grouped into 8 episodes; 6 episodes checkable against the saved measurements.

- E5 · 22 Oct 16:48–16:50 · Priority review: Power deviation (+19 dB) in C1; once, sub-threshold; also in: forensic case.
- E7 · 27 Oct 15:31 · Review: Power deviation (+21 dB) in C1; once, sub-threshold.
- E8 · 28 Oct 16:12–16:13 · Review: ICdetector forensic case in C2; once; also in: forensic case.
- Context: LTE band change E1, E3, E4; new item at a usual place E2; alternating PCI in the same cell E6.
- Radio records degraded states with no nearby ICdetector signals (4 out of service); they do not form an incident.

With the criteria implemented, no observation can be attributed to a fake base station or another attack; nor can it be stated that there has been no problem at all.

2. Incidents

Table 1. Incidents in chronological order. «Observation» summarises what ICdetector recorded; «Evidence» is CellCase's checks

Ep.	Date	Observation (ICdetector)	Evidence (CellCase)	Interpretation	Technical interest	Spoofing
E1	19 Oct 07:23	LTE band change (B3->B20) in C3; once, sub-threshold	Change visible in the rows: B3->B20 (C7->C3); No recurrence on other days	Consistent with band management; not conclusive	Context	Insufficient evidence
E2	21 Oct 03:11–03:12	New item at a usual place in C1; once; also in: Stable-Site	No CellCase check for this pattern; No recurrence on other days	No specific interpretation	Context	Insufficient evidence
E3	21 Oct 12:02–12:02	LTE band change (B7->B8) in C4; once, sub-threshold; also in: Stable-Site	Change visible in the rows: B7->B8 (C8->C4); No recurrence on other days	Consistent with band management; not conclusive	Context	Insufficient evidence
E4	21 Oct 12:33–12:33	LTE band change (B7->B20) in C5; once, sub-threshold; also in: Stable-Site	Change visible in the rows: B7->B20 (C2->C5); No recurrence on other days	Consistent with band management; not conclusive	Context	Insufficient evidence
E5	22 Oct 16:48–16:50	Power deviation (+19 dB) in C1; once, sub-threshold; also in: forensic case	Visible difference: -81 dBm against a median of -100 dBm (n=708, +19 dB); -81 dBm against a median of -100 dBm (n=708, +19 dB); Recurring: 2 days (E7)	Consistent with a change of position or band	Priority review	More data needed
E6	23 Oct 13:53–13:54	Alternating PCI in the same cell in C6; once, sub-threshold; also in: forensic case	Visible: repeated PCI alternation; No recurrence on other days	Several possible causes; not distinguishable without neighbours	Context	More data needed

Table 1. Incidents in chronological order. «Observation» summarises what ICdetector recorded; «Evidence» is …

Ep.	Date	Observation (ICdetector)	Evidence (CellCase)	Interpretation	Technical interest	Spoofing
E7	27 Oct 15:31	Power deviation (+21 dB) in C1; once, sub-threshold	Visible difference: -79 dBm against a median of -100 dBm (n=711, +21 dB); Recurring: 2 days (E5)	Consistent with a change of position or band	Review	Insufficient evidence
E8	28 Oct 16:12–16:13	ICdetector forensic case in C2; once; also in: forensic case	No CellCase check for this pattern; No recurrence on other days	No specific interpretation	Review	More data needed

Radio, in all incidents: no degraded events recorded (the continuity of the Radio log is not demonstrated).

Table 2. Short cell identifiers used in this report

Id.	Identity (MCC-MNC-TAC-CID-technology)	Id.	Identity
C1	`001-01-3001-1000070-LTE`	C5	`001-01-3002-1000543-LTE`
C2	`001-01-3002-1000542-LTE`	C6	`001-01-3003-1000953-LTE`
C3	`001-01-3001-1000069-LTE`	C7	`001-01-3001-1000980-LTE`
C4	`001-01-3002-1000545-LTE`	C8	`001-01-3002-1000544-LTE`

3. Incident details

E1 · 19 Oct 07:23 · LTE band change in C3 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 07:23:20 · history, C3: «[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (1805MHz->791MHz, B3->B20)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B3->B20 (C7->C3).
- A single observation at 07:23:20: its duration is unknown.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C3 (context): NOT_FOUND 73 rows.
- C3 is usual: 14 days as serving cell in the history.
- C3 has 36 of 54 consistent handovers in the Topology aggregate (no time per handover).

What it means and what it does not allow us to conclude

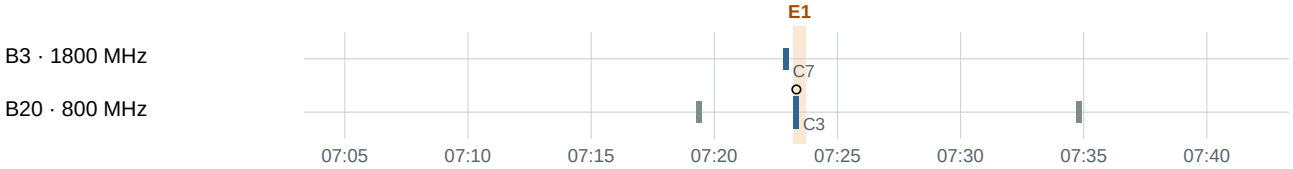
- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.

Figure 1. Band of the serving cell (E1)

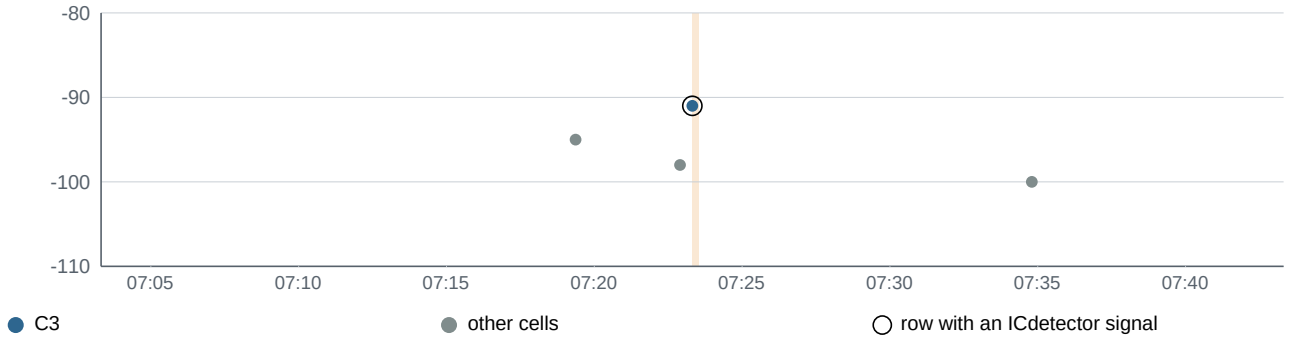
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 2. Serving cell power before and after (E1, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C1 (2 rows), C7 (1 row); during C3 (1 row); 30 min after C1 (1 row). References: icd31/SYNTHETIC_icd31.icdbackup › history.csv:748.

E2 · 21 Oct 03:11–03:12 · New item at a usual place in C1 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 03:11:54 · Stable-Site: «NOVEL_CELL»

What evidence supports it (CellCase checks)

- CellCase has no check of its own on the measurements for this type of pattern.
- 1 observation between 03:11:54 and 03:12:38; brief repetition. The interval does not demonstrate continuous activity.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C1 (context): NOT_FOUND 600 rows.
- C1 is usual: 15 days as serving cell in the history; up to 8 days in a Stable-Site key.
- C1 has 87 of 151 consistent handovers in the Topology aggregate (no time per handover).
- C1 is listed as KNOWN_ON_ROUTE in Geometry: a known route, not verified.

What it means and what it does not allow us to conclude

- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); CellCase has no check implemented for this pattern (not assessable does not mean not contradictory); it is a single observation.
- ICdetector's heuristics are not calibrated: their activation alone has no specific interpretation.
- No specific cause can be concluded.

What to check next

- Check whether that cell keeps appearing at that place over the following days (the same follow-up the plain-language report proposes).

Serving cells: 30 min before C1 (2 rows); during —; 30 min after C1 (1 row). References: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip!shadow_episodes.csv:4.

E3 · 21 Oct 12:02–12:02 · LTE band change in C4 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 12:02:12 · history, C4: «[sub-umbral] Downgrade de banda forzado (2620MHz->925MHz, B7->B8)» — explanatory translation: “[sub-threshold] Forced band downgrade (2620MHz->925MHz, B7->B8)”
- 12:02:12 · Stable-Site: «KNOWN_CELL»

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B7->B8 (C8->C4).
- 1 observation between 12:02:12 and 12:02:53; brief repetition. The interval does not demonstrate continuous activity.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- The sources of the episode derive from the same collector: they are not independent evidence of each other.
- Verification in public databases for C4 (context): NOT_FOUND 1 row.
- C4 appears in 1 row and on 1 day of the history (context; it does not change the priority).
- C4 has 1 of 2 consistent handovers in the Topology aggregate (no time per handover).

What it means and what it does not allow us to conclude

- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation; the other sources derive from the same collector and do not count as independent corroboration.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.

Figure 3. Band of the serving cell (E3)

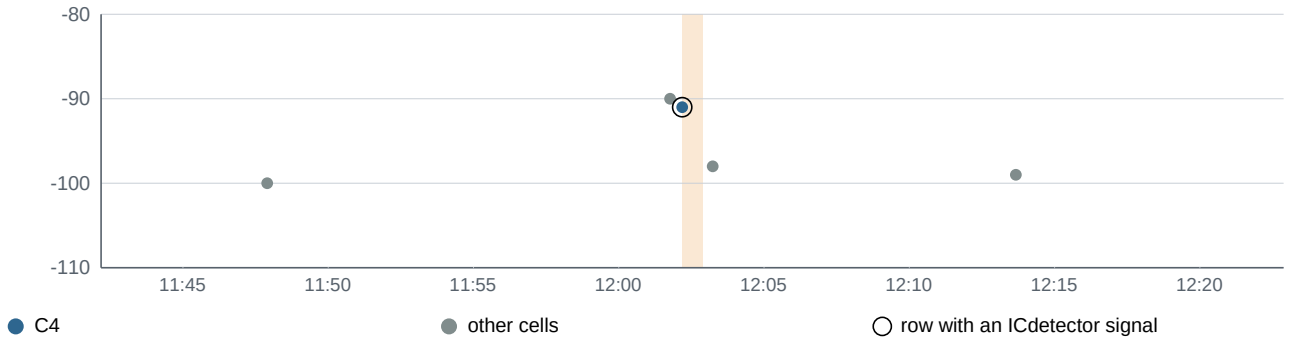
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 4. Serving cell power before and after (E3, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C3 (1 row), C1 (1 row), C8 (1 row); during C4 (1 row); 30 min after C1 (2 rows), C2 (1 row).
References: icd31/SYNTHETIC_icd31.icdbackup › history.csv:586; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip!shadow_episodes.csv:5.

E4 · 21 Oct 12:33–12:33 · LTE band change in C5 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 12:33:07 · history, C5: «[sub-umbral] Downgrade de banda forzado (2620MHz->791MHz, B7->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (2620MHz->791MHz, B7->B20)”
- 12:33:07 · Stable-Site: «NOVEL_CELL»

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B7->B20 (C2->C5).
- 1 observation between 12:33:07 and 12:33:49; brief repetition. The interval does not demonstrate continuous activity.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- The sources of the episode derive from the same collector: they are not independent evidence of each other.
- Verification in public databases for C5 (context): VERIFIED 1 row.
- C5 appears in 1 row and on 1 day of the history (context; it does not change the priority).
- C5 has 1 of 2 consistent handovers in the Topology aggregate (no time per handover).

What it means and what it does not allow us to conclude

- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation; the other sources derive from the same collector and do not count as independent corroboration.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.

Figure 5. Band of the serving cell (E4)

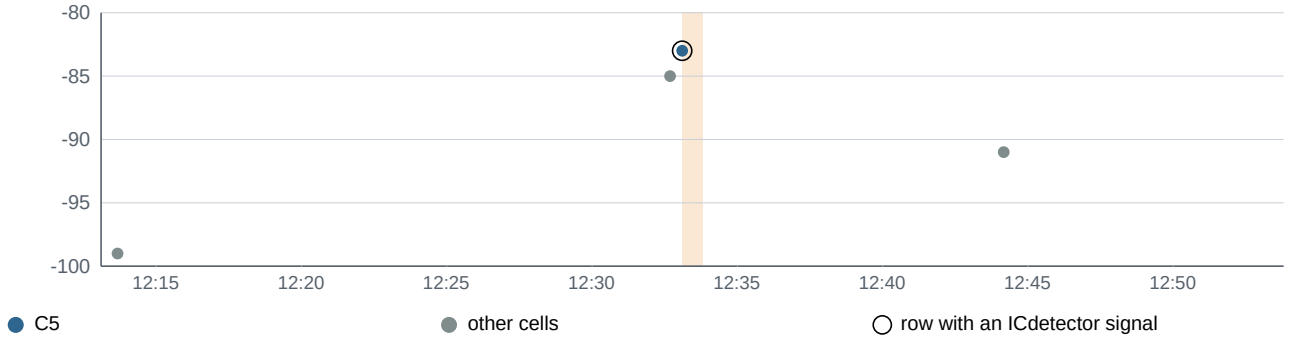
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 6. Serving cell power before and after (E4, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C1 (2 rows), C2 (1 row); during C5 (1 row); 30 min after C8 (2 rows). References: icd31/SYNTHETIC_icd31.icdbackup › history.csv:582; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip!shadow_episodes.csv:6.

E5 · 22 Oct 16:48–16:50 · Power deviation in C1 · Priority review

What was observed (recorded by ICdetector; verbatim quotes)

- 16:48:53 · history, C1: «[sub-umbral] Potencia anómala vs historial (+19dB) [confirmada 3/3]» — explanatory translation: “[sub-threshold] Potencia anómala vs historial (+19dB) [confirmada 3/3]”
- 16:48:53 · history, C1: «[sub-umbral] Potencia anómala vs historial (+19dB)» — explanatory translation: “[sub-threshold] Anomalous power vs history (+19dB)”
- 16:48:53 · forensic case ICD-2026-10-22-0001 (until 16:50:53)

What evidence supports it (CellCase checks)

- Against the reference for the same cell outside the episode: -81 dBm against a median of -100 dBm (n=708, +19 dB); -81 dBm against a median of -100 dBm (n=708, +19 dB).
- 1 observation between 16:48:53 and 16:50:53; brief repetition. The interval does not demonstrate continuous activity.
- The same pattern appears in C1 on 2 different days (also in E7). This is recurrence across days, not continuous activity.
- Radio: no degraded events recorded.
- The sources of the episode derive from the same collector: they are not independent evidence of each other.
- Verification in public databases for C1 (context): NOT_FOUND 600 rows.
- C1 is usual: 15 days as serving cell in the history; up to 8 days in a Stable-Site key.
- C1 has 87 of 151 consistent handovers in the Topology aggregate (no time per handover).
- C1 is listed as KNOWN_ON_ROUTE in Geometry: a known route, not verified.

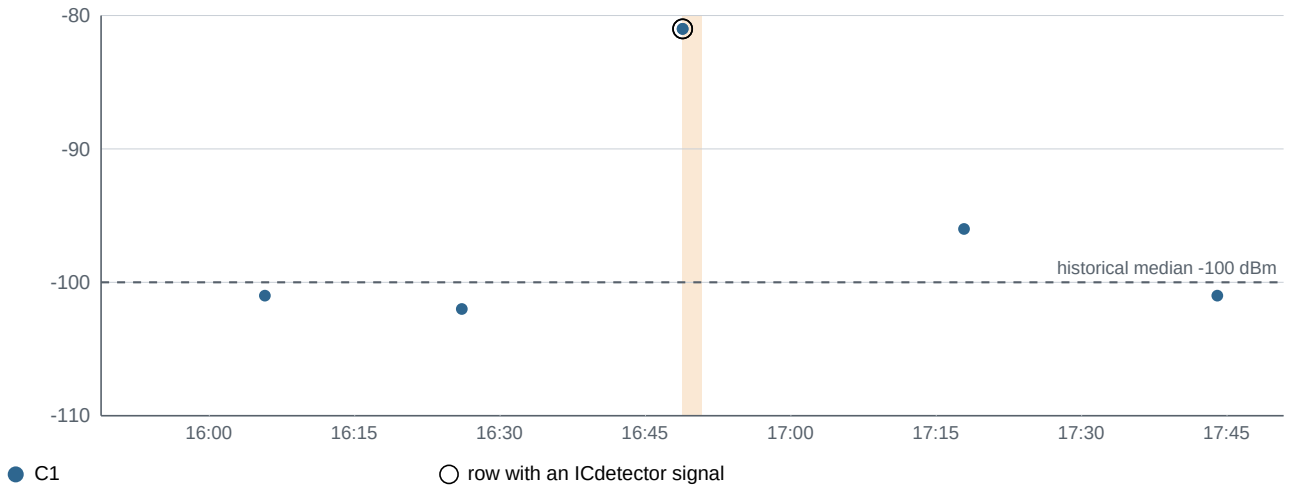
What it means and what it does not allow us to conclude

- Priority review: ICdetector opened a forensic case because of an alarm; the behaviour is confirmed in the measurements; it recurs on 2 days; it is a single observation; the other sources derive from the same collector and do not count as independent corroboration.
- Power changes with the position relative to the station (indoors or outdoors, height, obstacles) and with the band in use.
- The difference cannot be attributed to another transmitter: the reference mixes different positions and times.

What to check next

- Compare with captures at the exact same spot (the reference mixes different positions).
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Figure 7. Power of C1 and historical reference (E5)



Rows of C1 within ±1 h. Dashed line: median of the same cell outside the episode (n=708 rows, mixing positions and times). No interpolation.

Serving cells: 30 min before C1 (1 row), C7 (1 row); during C1 (2 rows); 30 min after 001-01-3001-1000071-LTE (1 row), C1 (1 row). References: icd31/SYNTHETIC_icd31.icdbackup › history.csv:493; icd31/SYNTHETIC_icd31.icdbackup › history.csv:494; icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip!timeline.csv:2; icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip!timeline.csv:6.

E6 · 23 Oct 13:53–13:54 · Alternating PCI in the same cell in C6 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 13:53:44 · history, C6: «[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)» — explanatory translation: “[sub-threshold] Unstable RF identity: same Cell ID alternating PCI on the same carrier (possible clone)”
- 13:53:44 · forensic case ICD-OBS-2026-10-23-0002 (until 13:54:14)
- 13:54:36 · history, C6: «[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)» — explanatory translation: “[sub-threshold] Unstable RF identity: same Cell ID alternating PCI on the same carrier (possible clone)”

What evidence supports it (CellCase checks)

- Within the same full cell identity and carrier, the rows show repeated PCI alternation: C6, EARFCN 1300: PCI 101 (13:53:20) -> PCI 102 (13:53:44) -> PCI 101 (13:54:10) -> PCI 102 (13:54:36). This describes what the file records; it does not prove that each row is a new modem reading.
- 1 observation between 13:53:44 and 13:54:36; brief repetition. The interval does not demonstrate continuous activity.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- The sources of the episode derive from the same collector: they are not independent evidence of each other.
- Verification in public databases for C6 (context): NOT_FOUND 4 rows.
- C6 appears in 4 rows and on 1 day of the history (context; it does not change the priority).
- C6 has 2 of 2 consistent handovers in the Topology aggregate (no time per handover).

What it means and what it does not allow us to conclude

- Context observation: there is an observation forensic case (it is not an alarm); ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation; the other sources derive from the same collector and do not count as independent corroboration.
- Several carriers or sectors can share part of the identity, and the modem may report data from the previous cell during a change.
- It may also be due to PCI re-planning by the operator.
- The alternation is visible in the data; without neighbours and diagnostics (forensic case) these explanations cannot be told apart.
- Those causes cannot be told apart without the neighbours and diagnostics of a forensic case.

What to check next

- If ICdetector opened a forensic case in that interval, export it: it includes neighbours and cycle-by-cycle diagnostics.
- Check whether the same PCI and EARFCN appear with another identity in other captures.

Figure 8. PCI and EARFCN of C6 (E6)

local time in Europe/Madrid (declared assumption)



History rows with the same full cell identity (MCC, MNC, area, Cell ID and technology) within ±30 min; one lane per PCI and carrier combination. Several lanes on the same carrier indicate PCI changes; the check table says whether it is a single PCI change, a PCI return or a repeated PCI alternation. Dotted mark: row with an ICdetector signal.

Serving cells: 30 min before C1 (1 row), C6 (1 row); during C6 (3 rows); 30 min after C1 (2 rows). References: icd31/SYNTHETIC_icd31.icdbackup › history.csv:429; icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip!timeline.csv:2; icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip!timeline.csv:3; icd31/SYNTHETIC_icd31.icdbackup › history.csv:427.

E7 · 27 Oct 15:31 · Power deviation in C1 · Review

What was observed (recorded by ICdetector; verbatim quotes)

- 15:31:20 · history, C1: «[sub-umbral] Potencia anómala vs historial (+21dB)» — explanatory translation: “[sub-threshold] Anomalous power vs history (+21dB)”

What evidence supports it (CellCase checks)

- Against the reference for the same cell outside the episode: -79 dBm against a median of -100 dBm (n=711, +21 dB).
- A single observation at 15:31:20: its duration is unknown.
- The same pattern appears in C1 on 2 different days (also in E5). This is recurrence across days, not continuous activity.
- Radio: no degraded events recorded.
- Verification in public databases for C1 (context): NOT_FOUND 600 rows.
- C1 is usual: 15 days as serving cell in the history; up to 8 days in a Stable-Site key.
- C1 has 87 of 151 consistent handovers in the Topology aggregate (no time per handover).
- C1 is listed as KNOWN_ON_ROUTE in Geometry: a known route, not verified.

What it means and what it does not allow us to conclude

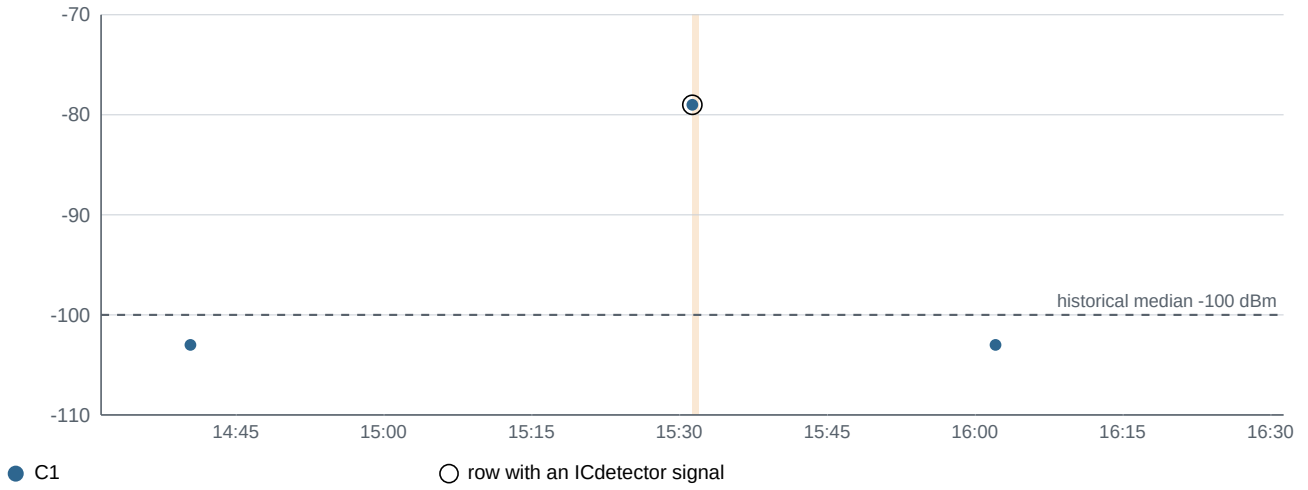
- Worth reviewing: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it recurs on 2 days; it is a single observation.
- Power changes with the position relative to the station (indoors or outdoors, height, obstacles) and with the band in use.
- The difference cannot be attributed to another transmitter: the reference mixes different positions and times.

What to check next

- Compare with captures at the exact same spot (the reference mixes different positions).
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Figure 9. Power of C1 and historical reference (E7)

dBm (getDbm())



Rows of C1 within ±1 h. Dashed line: median of the same cell outside the episode (n=711 rows, mixing positions and times). No interpolation.

Serving cells: 30 min before C7 (2 rows); during C1 (1 row); 30 min after 001-01-3001-1000071-LTE (1 row). References: icd31/SYNTHETIC_icd31.icdbackup › history.csv:169.

E8 · 28 Oct 16:12–16:13 · ICdetector forensic case in C2 · Review

What was observed (recorded by ICdetector; verbatim quotes)

- 16:12:00 · forensic case ICD-2026-10-28-0004 (until 16:13:30)

What evidence supports it (CellCase checks)

- CellCase has no check of its own on the measurements for this type of pattern.
- 1 observation between 16:12:00 and 16:13:30; brief repetition. The interval does not demonstrate continuous activity.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C2 (context): VERIFIED 32 rows.
- C2 is usual: 11 days as serving cell in the history.
- C2 has 31 of 60 consistent handovers in the Topology aggregate (no time per handover).

What it means and what it does not allow us to conclude

- Worth reviewing: ICdetector opened a forensic case because of an alarm; CellCase has no check implemented for this pattern (not assessable does not mean not contradictory); it is a single observation.
- ICdetector's heuristics are not calibrated: their activation alone has no specific interpretation.
- No specific cause can be concluded.

What to check next

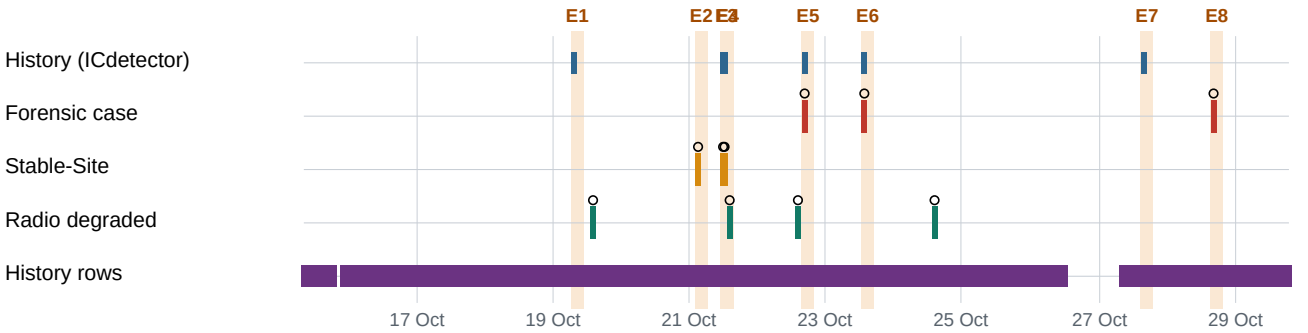
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Serving cells: 30 min before C3 (1 row); during 001-01-3001-1000071-LTE (1 row); 30 min after C8 (1 row). References: icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip!timeline.csv:2; icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip!timeline.csv:5.

4. Chronology

Figure 10. Signals by source over the period of the data

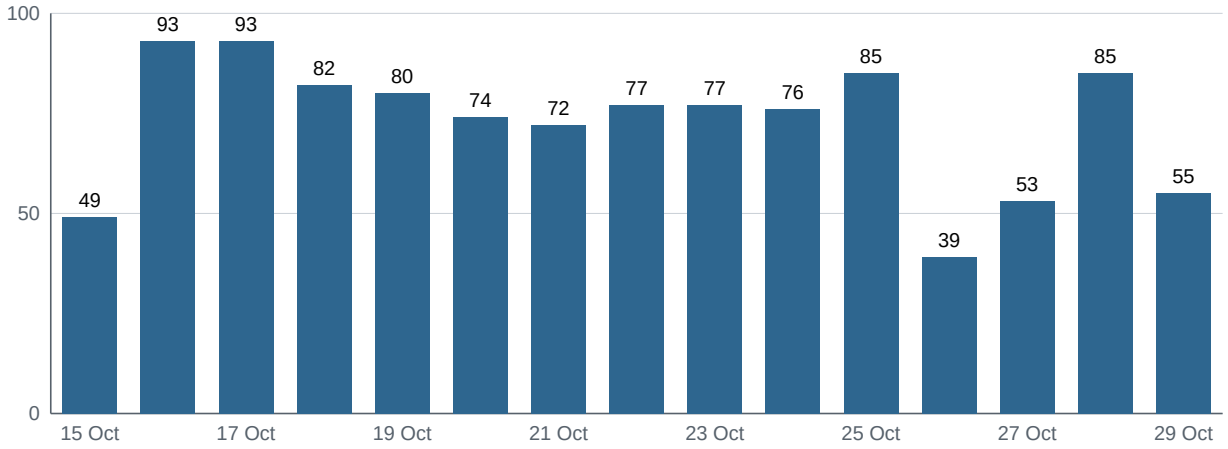
local time in Europe/Madrid (declared assumption)



Each mark is a record in its source (tall marks: not sub-threshold). «History rows» shows when there is sampling; the gaps are intervals with no rows. The bands are episodes or groups of episodes (G).

Figure 11. History rows by local date

Rows per day



The phone's local date; the height reflects ICdetector's sampling, not the phone's activity.

Figure 12. Service state changes recorded by Radio (whole period)

local time in Europe/Madrid (declared assumption)



Each mark is a recorded state change. Between marks it is unknown whether logging was continuous: the intervals are not drawn as sustained states. Source: Radio export (UTC).

Table 3. Intervals with no records of 60 minutes or more (the 10 longest)

From (last row before)	To (first row after)	Duration
2026-10-26 11:50:46 · icd31/SYNTHETIC_icd31.icdbackup › history.csv:195	2026-10-27 08:00:04 · icd31/SYNTHETIC_icd31.icdbackup › history.csv:194	20 h 09 min
2026-10-15 18:39:09 · icd31/SYNTHETIC_icd31.icdbackup › history.csv:1052	2026-10-15 21:54:00 · icd31/SYNTHETIC_icd31.icdbackup › history.csv:1051	3 h 14 min

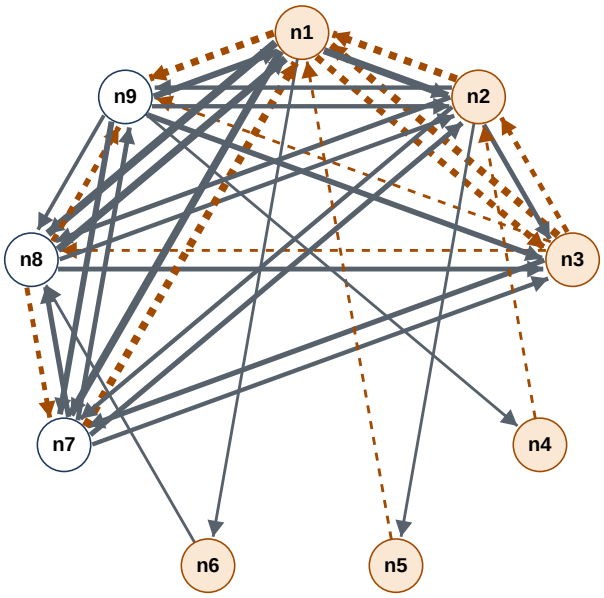
5. Network context

Topology, Geometry and Stable-Site are cumulative aggregates: they describe usual routes and places, not individual events, and they are not added to the history observations. Radio and forensic cases do have exact times.

Transitions between cells (Topology)

Content	36 directed routes, 232 cumulative handovers, generated 2026-10-29 19:40:11
Limit	Below the limit of 250 routes; this does not prove that it covers the whole campaign (retention, partial files).
Last state	12 routes with the last handover marked FAILED by the collector (it describes only the last one, not all).

Figure 13. Transitions observed between cells (Topology aggregate)



n1 = 001-01-3001-1000070-LTE · 151 handovers · in an episode
n2 = 001-01-3002-1000542-LTE · 60 handovers · in an episode
n3 = 001-01-3001-1000069-LTE · 54 handovers · in an episode
n4 = 001-01-3002-1000545-LTE · 2 handovers · in an episode
n5 = 001-01-3002-1000543-LTE · 2 handovers · in an episode
n6 = 001-01-3003-1000953-LTE · 2 handovers · in an episode
n7 = 001-01-3001-1000980-LTE · 72 handovers
n8 = 001-01-3001-1000071-LTE · 65 handovers
n9 = 001-01-3002-1000544-LTE · 56 handovers

Only routes present in transitions.csv are drawn; the thickness grows with the cumulative number of handovers (logarithmic scale). Dashed line: the last handover on that route was FAILED. Highlighted nodes: cells that appear in some episode. Selection: cells from episodes (up to 6) and then those with the most handovers, up to 14. Omitted: 0 cells and 0 routes. The aggregate does not say when each handover happened.

Service state (Radio)

Content	11 state changes between 2026-10-18 11:00:00 and 2026-10-24 14:44:00
States	IN_SERVICE 6; OUT_OF_SERVICE 4; POWER_OFF 1
Operator different from the SIM without roaming	0
Degraded states with no other signals	4 out of service (they do not form an episode)
Consistency with the history	See «Agreements and differences between sources» (X-SVC).

Forensic cases

Table 4. Imported forensic cases (ICdetector's assessments)

Case	State / origin	Window	Samples	Cell	Confirmed
ICD-2026-10-22-0001	READY / ALARM (alarm)	22 Oct 16:48 – 16:50	5	`001-01-3001-1000070-LTE`	yes
ICD-OBS-2026-10-23-0002	INTERRUPTED / TRUST_CONTRADICTION (observation, not an alarm)	23 Oct 13:53 – 13:54	2	`001-01-3003-1000953-LTE`	no
ICD-2026-10-26-0003	CAPTURING / ALARM (alarm)	? – ?	0	`001-01-3001-1000980-LTE`	no
ICD-2026-10-28-0004	READY / ALARM (alarm)	28 Oct 16:12 – 16:13	4	`001-01-3002-1000542-LTE`	no

Usual places (Stable-Site)

Each place is recorded under up to four overlapping site keys, so there is no single total of observations: they are shown per key. The keys are hashed summaries of the position; CellCase does not turn them into coordinates.

Table 5. Site keys with the most observations (at most 10 of 3)

Key	State	Serving days	Static days	Serving obs.	Cells
`a1b2c3d4e5…`	ACTIVE	8	6	90	3
`0f1e2d3c4b…`	SHADOW_READY	6	4	60	2
`ffeeddccbb…`	LEARNING	4	2	30	1

Shadow episodes: 5, of which 3 were recorded as a possible novelty (collector result; in 2.10.5 they do not block anything).

Mobility (Geometry)

Content	9 cells, 36 routes and 6 trips; no coordinates
Familiarity	UNKNOWN_ON_ROUTE 5; KNOWN_ON_ROUTE 3; OBSERVED_ON_ROUTE 1. KNOWN_ON_ROUTE means a known route, neither trusted nor verified.
Trips	1 open; 2 closed that do not meet the validity rule (discarded by ICdetector).
Limit	Below the limit of 1,000; it does not prove complete coverage.

6. Agreements and differences between sources

Compares the sources of this case with each other, using full cell identities and times with a single instant. Each result is classified as an agreement, a difference, complementary information or not comparable. An absence from a file does not prove a contradiction, and a difference does not prove a network anomaly or an IMSI catcher. Almost all sources come from ICdetector: an agreement between them checks that the files are consistent; it is not an independent confirmation. The cross-checks are context and do not change the assessment.

Results: 12 agreements, 2 differences, 13 complementary and 7 not comparable. The totals count check results (a row, a cell, a route or a Stable-Site episode compared), not incidents or independent evidence.

Summary by check

Table 6. Checks cross/3: 23 groups covering 34 results; each result is in a single group. Examples: the first results of each group, with their reference (file, entry and physical line).

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-SVC Service state History / Radio	Agreement 1 result · 377 items compared · 18 Oct 11:05 – 23 Oct 10:57	Both sources record the same value at the compared moments. • 377 rows. state: History IN_SERVICE · Radio IN_SERVICE (377). 18 Oct 11:05 – 23 Oct 10:57. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:814; icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:2	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated.
X-SVC Service state History / Radio	Difference 1 result · 90 items compared · 23 Oct 11:18 – 24 Oct 14:32	The sources record different values for the same entity at comparable moments. • 90 rows. state: History IN_SERVICE · Radio POWER_OFF (90). 23 Oct 11:18 – 24 Oct 14:32. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:435; icd31/SYNTHETIC_icd31.icdbackup › history.csv:436	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated. A difference does not prove a network anomaly or an IMSI catcher. Next step: review the cited rows in the original files.
X-SVC Service state History / Radio	Not comparable 1 result · 623 items compared · 15 Oct 08:00 – 29 Oct 18:48	Rows before the first Radio event or after the last one: Radio does not say until when it covers. • 623 rows. 15 Oct 08:00 – 29 Oct 18:48. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:1089; icd31/SYNTHETIC_icd31.icdbackup › history.csv:1090	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated. Next step: provide a Radio export that covers that period.
X-CELL Serving cells History / Topology, Geometry	Agreement 1 result · 9 items compared · 9 entities	The cells appear as serving cells in both sources. • 9 cells: C3, C1 and 7 more. C3: rows: History 87. and 7 more. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:2; icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:3	Both sources come from ICdetector.
X-CELL Serving cells History / Stable-Site	Agreement 1 result · 3 items compared · 3 entities	The cells appear as serving cells in both sources. • 3 cells: C1, C9 and 1 more. C1: rows: History 712. and 5 more. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:2; icd31/SYNTHETIC_icd31.icdbackup › history.csv:3	Both sources come from ICdetector.

Table 6. Checks cross/3: 23 groups covering 34 results; each result is in a single group. Examples: the first res…

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-CELL Serving cells History / Stable-Site	Complementary 1 result · 6 items compared · 6 entities	History cells that the aggregate does not contain: the aggregate only records part of the data (recent routes, places with a precise position) and may be from another time. • 6 cells: C3, C2 and 4 more. C3: rows: History 87. and 5 more. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:4; icd31/SYNTHETIC_icd31.icdbackup › history.csv:13	Both sources come from ICdetector.
X-ROUTE Cell changes and routes History / Topology, Geometry	Agreement 1 result · 14 items compared · 14 entities · 16 Oct 09:17 – 28 Oct 17:41	The cell changes in the history exist as routes in the aggregate. Counters are not added up: the history is sampled and the aggregate is cumulative. • 14 routes: C3 -> C7, C1 -> C7 and 12 more. C3 -> C7: changes: History 2 · Topology 5. and 7 more. 16 Oct 09:17 – 28 Oct 17:41. window ±10 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_edges.csv:2; icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_edges.csv:4	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.
X-ROUTE Cell changes and routes History / Topology, Geometry	Not comparable 1 result · 1 item compared · 1 entity · 21 Oct 12:03	The route is not in the aggregate; the history is sampled and an intermediate cell may be missing. • 1 route: C4 -> C1. C4 -> C1: changes: History 1. 21 Oct 12:03. window ±10 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:585; icd31/SYNTHETIC_icd31.icdbackup › history.csv:586	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.
X-AGG Route counters Topology / Geometry	Agreement 1 result · 36 items compared · 36 entities · 29 Oct 19:40	Both sources record the same value at the compared moments. • 36 routes: C3 -> C1, C3 -> C9 and 34 more. C3 -> C1: handovers: Topology 16 · Geometry 16. and 7 more. 29 Oct 19:40. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_edges.csv:2; icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_edges.csv:3	Counters from the same ICdetector database at different times.
X-SHADOW Stable-Site episode Stable-Site / History	Agreement 5 results · 6 items compared · 9 entities · 19 Oct 07:23 – 21 Oct 12:33	The history shows the candidate cell as serving within the window of the Stable-Site episode (margin ±5 s). It does not prove that it is the same measurement. • 1 row in the window: C3, ep-000. serving cell: Stable-Site C3 · History C3. 19 Oct 07:23 – 19 Oct 07:24. window ±5 s. • 1 row in the window: C1, ep-003. serving cell: Stable-Site C1 · History C1. 19 Oct 18:43 – 19 Oct 18:44. window ±5 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:748; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:2; icd31/SYNTHETIC_icd31.icdbackup › history.csv:712 (and 3 more results of the same type)	Both sources come from ICdetector.

Table 6. Checks cross/3: 23 groups covering 34 results; each result is in a single group. Examples: the first res…

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-SHADOW Stable-Site episode Stable-Site / Topology	Agreement 1 result · 1 item compared · 2 entities · 19 Oct 07:23 – 19 Oct 07:24	The route recorded its last change within the window of the Stable-Site episode (margin ±1 min). It is compatible with the same cell change, but coinciding within the margin does not prove that it is exactly the same event. • 1 episode: C7 -> C3, ep-000. handovers: Topology 3. and 2 more. 19 Oct 07:23 – 19 Oct 07:24. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:2; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:18	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.
X-SHADOW Stable-Site episode Stable-Site / Topology	Complementary 1 result · 1 item compared · 2 entities · 21 Oct 03:11 – 21 Oct 03:12	The route exists in Topology, but its last recorded change is from another time: it cannot be said whether it includes the one in this episode. • 1 episode: C7 -> C1, ep-004. handovers: Topology 19. and 2 more. 21 Oct 03:11 – 21 Oct 03:12. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:4; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:19	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.
X-SHADOW Stable-Site episode Stable-Site / Topology	Not comparable 2 results · 2 items compared · 2 entities · 19 Oct 18:43 – 21 Oct 12:03	Incomplete identities or unknown technology: not used to relate sources (the PCI does not identify a cell). • 1 episode: ep-003. 19 Oct 18:43 – 19 Oct 18:44. window ±1 min. • 1 episode: ep-001. 21 Oct 12:02 – 21 Oct 12:03. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:3; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:5	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.
X-SHADOW Stable-Site episode Stable-Site / Topology	Not comparable 1 result · 1 item compared · 2 entities · 21 Oct 12:33 – 21 Oct 12:33	The route is not in the export; an absence does not prove that it did not happen. • 1 episode: C7 -> C5, ep-002. route: Stable-Site C7 -> C5. 21 Oct 12:33 – 21 Oct 12:33. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:6	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation. Next step: export the aggregate again after those dates.
X-FOR Serving cell of the case Forensic case / History	Agreement 1 result · 2 items compared · 1 entity · 22 Oct 16:48 – 22 Oct 16:50	Both sources record the same value at the compared moments. • 2 rows or samples: ICD-2026-10-22-0001. serving cell: Forensic case C1 · History C1 (2). 22 Oct 16:48 – 22 Oct 16:50. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip! cells.csv:2; icd31/SYNTHETIC_icd31.icdbackup › history.csv:493	The same measurement recorded in two files: it checks that the files are consistent; it is not an independent confirmation.
X-FOR Serving cell of the case Forensic case / History	Difference 1 result · 1 item compared · 1 entity · 23 Oct 13:53 – 23 Oct 13:54	The sources record different values for the same entity at comparable moments. • 1 row or sample: ICD-OBS-2026-10-23-0002. C6: PCI: Forensic case 101 · History 102. 23 Oct 13:53 – 23 Oct 13:54. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip! cells.csv:2; icd31/SYNTHETIC_icd31.icdbackup › history.csv:429	The same measurement recorded in two files: it checks that the files are consistent; it is not an independent confirmation. A difference does not prove a network anomaly or an IMSI catcher. Next step: review the cited rows in the original files.

Table 6. Checks cross/3: 23 groups covering 34 results; each result is in a single group. Examples: the first res…

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-FOR Serving cell of the case Forensic case / History	Complementary 3 results · 11 items compared · 3 entities · 22 Oct 16:48 – 28 Oct 16:13	The forensic case includes neighbouring cells that the history does not keep. • 5 rows or samples: ICD-2026-10-22-0001. neighbours: Forensic case 5. 22 Oct 16:48 – 22 Oct 16:50. window ±3 s. • 2 rows or samples: ICD-OBS-2026-10-23-0002. neighbours: Forensic case 2. 23 Oct 13:53 – 23 Oct 13:54. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip! cells.csv:3; icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip! cells.csv:3 (and 1 more result of the same type)	Both sources come from ICdetector.
X-FOR Serving cell of the case Forensic case / History	Not comparable 1 result · 1 item compared · 1 entity · 28 Oct 16:12 – 28 Oct 16:13	There are no history rows in the window: the history is sampled. • 1 row or sample: ICD-2026-10-28-0004. rows: Forensic case 1 · History 4. 28 Oct 16:12 – 28 Oct 16:13. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:91	The same measurement recorded in two files: it checks that the files are consistent; it is not an independent confirmation. Next step: if ICdetector opened a forensic case at that time, export it.
X-FOR-RAD Service state during the case Forensic case / Radio	Complementary 2 results · 0 items compared · 2 entities · 22 Oct 16:48 – 23 Oct 13:54	Radio recorded no state changes during the case window. • 0 events in the window: ICD-2026-10-22-0001. 22 Oct 16:48 – 22 Oct 16:50. window ±1 min. • 0 events in the window: ICD-OBS-2026-10-23-0002. 23 Oct 13:53 – 23 Oct 13:54. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:8; icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:10	Different families (phone state and collector): coinciding in time is context, not a cause.
X-FOR-RAD Service state during the case Forensic case / Radio	Not comparable 1 result · 0 items compared · 1 entity · 28 Oct 16:12 – 28 Oct 16:13	Rows before the first Radio event or after the last one: Radio does not say until when it covers. • 0 events in the window: ICD-2026-10-28-0004. 28 Oct 16:12 – 28 Oct 16:13. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:12	Different families (phone state and collector): coinciding in time is context, not a cause. Next step: provide a Radio export that covers that period.
X-FOR-AGG Case cell in aggregates Forensic case / Topology, Geometry, Stable-Site	Complementary 2 results · 2 items compared · 4 entities · 22 Oct 16:48 – 22 Oct 16:50	The case cell appears in the aggregate; this does not show that the route was used during the case. • 1 cell: C7, ICD-2026-10-26-0003. C7: handovers: Topology 72. and 2 more. • 1 cell: C1, ICD-2026-10-22-0001. C1: handovers: Topology 151. and 2 more. 22 Oct 16:48 – 22 Oct 16:50. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:5; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! site_cells.csv:3; icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:3	Both sources come from ICdetector.

Table 6. Checks cross/3: 23 groups covering 34 results; each result is in a single group. Examples: the first res…

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-FOR-AGG Case cell in aggregates Forensic case / Topology, Geometry	Complementary 2 results · 2 items compared · 4 entities · 23 Oct 13:53 – 28 Oct 16:13	The case cell appears in the aggregate; this does not show that the route was used during the case. • 1 cell: C6, ICD-OBS-2026-10-23-0002. C6: handovers: Topology 2. and 1 more. 23 Oct 13:53 – 23 Oct 13:54. • 1 cell: C2, ICD-2026-10-28-0004. C2: handovers: Topology 60. and 1 more. 28 Oct 16:12 – 28 Oct 16:13. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:10; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:12; icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:6	Both sources come from ICdetector.
X-FOR-AGG Case cell in aggregates Forensic case / Stable-Site	Complementary 2 results · 2 items compared · 4 entities · 23 Oct 13:53 – 28 Oct 16:13	The case cell does not appear in the aggregate. • 1 cell: C6, ICD-OBS-2026-10-23-0002. 23 Oct 13:53 – 23 Oct 13:54. • 1 cell: C2, ICD-2026-10-28-0004. 28 Oct 16:12 – 28 Oct 16:13.	Both sources come from ICdetector.

Differences and cross-checks linked to the episodes

Table 7. Differences and results linked to episodes, grouped by episode and type so that the same context is not repeated. Their limitations are in the summary.

Episodes	Comparison	Result	Details and references
E1	X-SHADOW Stable-Site episode Stable-Site / History	Agreement	The history shows the candidate cell as serving within the window of the Stable-Site episode (margin ±5 s). It does not prove that it is the same measurement. 1 row in the window: C3, ep-000. serving cell: Stable-Site C3 · History C3. 19 Oct 07:23 – 19 Oct 07:24. window ±5 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:748; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:2
E1	X-SHADOW Stable-Site episode Stable-Site / Topology	Agreement	The route recorded its last change within the window of the Stable-Site episode (margin ±1 min). It is compatible with the same cell change, but coinciding within the margin does not prove that it is exactly the same event. 1 episode: C7 -> C3, ep-000. handovers: Topology 3. time: Stable-Site 19 Oct 07:23 · Topology 19 Oct 07:23. route: Stable-Site C7 -> C3 · Topology C7 -> C3. 19 Oct 07:23 – 19 Oct 07:24. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:2; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:18
E2	X-SHADOW Stable-Site episode Stable-Site / History	Agreement	The history shows the candidate cell as serving within the window of the Stable-Site episode (margin ±5 s). It does not prove that it is the same measurement. 1 row in the window: C1, ep-004. serving cell: Stable-Site C1 · History C1. 21 Oct 03:11 – 21 Oct 03:12. window ±5 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:612; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:4
E2	X-SHADOW Stable-Site episode Stable-Site / Topology	Complementary	The route exists in Topology, but its last recorded change is from another time: it cannot be said whether it includes the one in this episode. 1 episode: C7 -> C1, ep-004. handovers: Topology 19. time: Stable-Site 21 Oct 03:11 · Topology 26 Oct 10:19. route: Stable-Site C7 -> C1 · Topology C7 -> C1. 21 Oct 03:11 – 21 Oct 03:12. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:4; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:19

Table 7. Differences and results linked to episodes, grouped by episode and type so that the same context is no…

Episodes	Comparison	Result	Details and references
E3	X-SHADOW Stable-Site episode Stable-Site / History	Agreement	The history shows the candidate cell as serving within the window of the Stable-Site episode (margin ±5 s). It does not prove that it is the same measurement. 2 rows in the window: C4, ep-001. serving cell: Stable-Site C4 · History C1. serving cell: Stable-Site C4 · History C4. 21 Oct 12:02 – 21 Oct 12:03. window ±5 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:585; icd31/SYNTHETIC_icd31.icdbackup › history.csv:586; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:5
E3	X-SHADOW Stable-Site episode Stable-Site / Topology	Not comparable	Incomplete identities or unknown technology: not used to relate sources (the PCI does not identify a cell). 1 episode: ep-001. 21 Oct 12:02 – 21 Oct 12:03. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:5
E4	X-SHADOW Stable-Site episode Stable-Site / History	Agreement	The history shows the candidate cell as serving within the window of the Stable-Site episode (margin ±5 s). It does not prove that it is the same measurement. 1 row in the window: C5, ep-002. serving cell: Stable-Site C5 · History C5. 21 Oct 12:33 – 21 Oct 12:33. window ±5 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:582; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:6
E4	X-SHADOW Stable-Site episode Stable-Site / Topology	Not comparable	The route is not in the export; an absence does not prove that it did not happen. 1 episode: C7 -> C5, ep-002. route: Stable-Site C7 -> C5. 21 Oct 12:33 – 21 Oct 12:33. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! shadow_episodes.csv:6
E5	X-FOR Serving cell of the case Forensic case / History	Agreement	Both sources record the same value at the compared moments. 2 rows or samples: ICD-2026-10-22-0001. serving cell: Forensic case C1 · History C1 (2). 22 Oct 16:48 – 22 Oct 16:50. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip! cells.csv:2; icd31/SYNTHETIC_icd31.icdbackup › history.csv:493; icd31/SYNTHETIC_icd31.icdbackup › history.csv:494
E5	X-FOR Serving cell of the case Forensic case / History	Complementary	The forensic case includes neighbouring cells that the history does not keep. 5 rows or samples: ICD-2026-10-22-0001. neighbours: Forensic case 5. 22 Oct 16:48 – 22 Oct 16:50. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip! cells.csv:3
E5	X-FOR-RAD Service state during the case Forensic case / Radio	Complementary	Radio recorded no state changes during the case window. 0 events in the window: ICD-2026-10-22-0001. 22 Oct 16:48 – 22 Oct 16:50. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:8
E5	X-FOR-AGG Case cell in aggregates Forensic case / Topology, Geometry, Stable-Site	Complementary	The case cell appears in the aggregate; this does not show that the route was used during the case. 1 cell: C1, ICD-2026-10-22-0001. C1: handovers: Topology 151. C1: days as serving cell (max. per key): Stable-Site 8. C1: familiarity: Geometry KNOWN_ON_ROUTE. 22 Oct 16:48 – 22 Oct 16:50. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:3; icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip! site_cells.csv:2; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:2
E6	X-SVC Service state History / Radio	Difference	The sources record different values for the same entity at comparable moments. 90 rows. state: History IN_SERVICE · Radio POWER_OFF (90). 23 Oct 11:18 – 24 Oct 14:32. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:435; icd31/SYNTHETIC_icd31.icdbackup › history.csv:436; icd31/SYNTHETIC_icd31.icdbackup › history.csv:437 …

Table 7. Differences and results linked to episodes, grouped by episode and type so that the same context is no…

Episodes	Comparison	Result	Details and references
E6	X-FOR Serving cell of the case Forensic case / History	Difference	The sources record different values for the same entity at comparable moments. 1 row or sample: ICD-OBS-2026-10-23-0002. C6: PCI: Forensic case 101 · History 102. 23 Oct 13:53 – 23 Oct 13:54. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip! cells.csv:2; icd31/SYNTHETIC_icd31.icdbackup › history.csv:429
E6	X-FOR Serving cell of the case Forensic case / History	Complementary	The forensic case includes neighbouring cells that the history does not keep. 2 rows or samples: ICD-OBS-2026-10-23-0002. neighbours: Forensic case 2. 23 Oct 13:53 – 23 Oct 13:54. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip! cells.csv:3
E6	X-FOR-RAD Service state during the case Forensic case / Radio	Complementary	Radio recorded no state changes during the case window. 0 events in the window: ICD-OBS-2026-10-23-0002. 23 Oct 13:53 – 23 Oct 13:54. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:10
E6	X-FOR-AGG Case cell in aggregates Forensic case / Topology, Geometry	Complementary	The case cell appears in the aggregate; this does not show that the route was used during the case. 1 cell: C6, ICD-OBS-2026-10-23-0002. C6: handovers: Topology 2. C6: familiarity: Geometry UNKNOWN_ON_ROUTE. 23 Oct 13:53 – 23 Oct 13:54. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:10; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:12
E6	X-FOR-AGG Case cell in aggregates Forensic case / Stable-Site	Complementary	The case cell does not appear in the aggregate. 1 cell: C6, ICD-OBS-2026-10-23-0002. 23 Oct 13:53 – 23 Oct 13:54.
E8	X-FOR Serving cell of the case Forensic case / History	Complementary	The forensic case includes neighbouring cells that the history does not keep. 4 rows or samples: ICD-2026-10-28-0004. neighbours: Forensic case 4. 28 Oct 16:12 – 28 Oct 16:13. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip! cells.csv:3
E8	X-FOR Serving cell of the case Forensic case / History	Not comparable	There are no history rows in the window: the history is sampled. 1 row or sample: ICD-2026-10-28-0004. rows: Forensic case 1 · History 4. 28 Oct 16:12 – 28 Oct 16:13. window ±3 s. Ref.: icd31/SYNTHETIC_icd31.icdbackup › history.csv:91
E8	X-FOR-RAD Service state during the case Forensic case / Radio	Not comparable	Rows before the first Radio event or after the last one: Radio does not say until when it covers. 0 events in the window: ICD-2026-10-28-0004. 28 Oct 16:12 – 28 Oct 16:13. window ±1 min. Ref.: icd31/SYNTHETIC_icd31.icdbackup › service_state.csv:12
E8	X-FOR-AGG Case cell in aggregates Forensic case / Topology, Geometry	Complementary	The case cell appears in the aggregate; this does not show that the route was used during the case. 1 cell: C2, ICD-2026-10-28-0004. C2: handovers: Topology 60. C2: familiarity: Geometry UNKNOWN_ON_ROUTE. 28 Oct 16:12 – 28 Oct 16:13. Ref.: icd31/SYNTHETIC_icd31.icdbackup › geometry.zip! geometry_cells.csv:6; icd31/SYNTHETIC_icd31.icdbackup › topology.zip! transitions.csv:5
E8	X-FOR-AGG Case cell in aggregates Forensic case / Stable-Site	Complementary	The case cell does not appear in the aggregate. 1 cell: C2, ICD-2026-10-28-0004. 28 Oct 16:12 – 28 Oct 16:13.

The full list of the 34 results is not included: it is an optional annex, off by default. The summary totals are the same.

New short identifiers in this table: C9 = 001-01-3001-1000071-LTE.

7. ICdetector local profile (shadow mode)

ICdetector 3.0 learns on each phone how the serving signal of each cell usually looks (level in dBm and, on LTE/NR, RSRQ quality) and records, for every history row, how far the reading is from usual: deviation = |value - mean| / spread. The largest deviation sets the level: below 2, NONE; below 3, LOW; below 4, MODERATE; from 4, HIGH. When it cannot compare (no or little-known reference, another phone model, another technology…) it abstains (ABSTAIN) and says why.

It is context, as in ICdetector: it runs in shadow mode, its thresholds are not calibrated, a deviation is not a probability and HIGH is not an alert. CellCase shows and summarises it, but does not use it for episodes, priorities or the assessment.

Rows evaluated	582 of 1,090 (813 recorded with ICdetector 3.0)
Levels	normal (NONE) 473 · low (LOW) 68 · high (HIGH) 1 · not compared (ABSTAIN) 40
Abstentions	cell without reference 22; reference still learning 18
Mode	shadow 582
Algorithm version (rows)	2 (582)
Held out for evaluation (not learned)	64
Evaluation period	18 Oct 11:05 – 26 Oct 11:40
References in the backup	mature 6

Table 8. Local profile in the rows of each episode (profile/1). Deviations are computed by ICdetector; CellCase does not recompute them.

Episode	Evaluated / rows	Levels	Largest deviation (reading and reference)
E1	1 / 1	normal (NONE) 1	normal (NONE), deviation 0.33, 19 Oct 07:23 (icd31/SYNTHETIC_icd31.icdbackup › history.csv:748) Values: SIGNAL_DBM -91 vs -92.000±3.000 z=0.333; SIGNAL_RSRQ -11 vs -10.500±1.500 z=0.333
E3	1 / 1	normal (NONE) 1	normal (NONE), deviation 1.00, 21 Oct 12:02 (icd31/SYNTHETIC_icd31.icdbackup › history.csv:586) Values: SIGNAL_DBM -91 vs -88.000±3.000 z=1.000; SIGNAL_RSRQ -10 vs -10.500±1.500 z=0.333
E4	1 / 1	normal (NONE) 1	normal (NONE), deviation 0.67, 21 Oct 12:33 (icd31/SYNTHETIC_icd31.icdbackup › history.csv:582) Values: SIGNAL_DBM -83 vs -85.000±3.000 z=0.667; SIGNAL_RSRQ -12 vs -10.500±1.500 z=1.000
E5	1 / 2	high (HIGH) 1	high (HIGH), deviation 6.33, 22 Oct 16:48 (icd31/SYNTHETIC_icd31.icdbackup › history.csv:494) Values: SIGNAL_DBM -81 vs -100.000±3.000 z=6.333; SIGNAL_RSRQ -12 vs -10.500±1.500 z=1.000
E6	2 / 3	normal (NONE) 2	normal (NONE), deviation 1.67, 23 Oct 13:54 (icd31/SYNTHETIC_icd31.icdbackup › history.csv:428) Values: SIGNAL_DBM -113 vs -108.000±3.000 z=1.667; SIGNAL_RSRQ -12 vs -10.500±1.500 z=1.000
E7	0 / 1	—	—
E8	0 / 1	—	—

Outside the episodes there are no rows with level MODERATE or HIGH.

Limitations: the profile looks at the same signal as ICdetector's power rule (H13), so agreeing with it is not independent confirmation. It does not resist evasion: a transmitter present while it was learning, or one imitating the usual signal, looks normal to it. A reference belongs to a cell identity, not to a place: a legitimate change (re-tilted antenna, another room) gives high deviations until the reference expires. Rows before 3.0 have no profile, and held-out rows were not learned.

8. Cell churn (H17, shadow mode, not calibrated)

What it measures: how many distinct cell identities have served the phone in the previous 15 minutes (ChurnCells15m) and how many serving changes there have been (ChurnChanges15m), with the motion state (ChurnMotion: still, moving or unknown). The count restarts after a loss of service or a gap of more than 2 minutes.

What it means in ICdetector 3.1.0: shadow mode. It is recorded, not evaluated, subtracts no points and has no thresholds; ICdetector will calibrate them with campaign data. A high value while still is something to look at, not an alarm. CellCase describes it and does not use it for episodes, priorities or the assessment.

Rows with cell churn 193 (rows from ICdetector 3.1.0 or later: 193; history rows: 1,090)
Period with cell churn 27 Oct 08:00 – 29 Oct 18:48

Table 9. Cell churn distribution by motion state (churn/1). Nearest-rank percentiles; descriptive data, no thresholds.

Motion	Rows	Cells in 15 min (median · P90 · max)	Changes in 15 min (median · P90 · max)
still	123	1 · 2 · 2	1 · 2 · 2
moving	51	3 · 6 · 7	2 · 9 · 11
unknown	19	3 · 4 · 4	3 · 4 · 4

Table 10. High-value periods: consecutive rows (15 min apart or less) with at least 4 changes in 15 min, this case's P90. It is a reference from the history itself, not an ICdetector threshold; up to five periods, from most to fewest changes.

Period	Rows	Peak	Motion	Row with most changes
28 Oct 17:09 – 28 Oct 17:39	12	11 changes · 7 cells	moving	icd31/SYNTHETIC_icd31.icdbackup › history.csv:81
27 Oct 15:41 – 27 Oct 15:41	1	4 changes · 4 cells	unknown	icd31/SYNTHETIC_icd31.icdbackup › history.csv:168
27 Oct 22:40 – 27 Oct 22:40	1	4 changes · 4 cells	moving	icd31/SYNTHETIC_icd31.icdbackup › history.csv:145
28 Oct 00:16 – 28 Oct 00:16	1	4 changes · 4 cells	unknown	icd31/SYNTHETIC_icd31.icdbackup › history.csv:141
28 Oct 05:13 – 28 Oct 05:13	1	4 changes · 4 cells	moving	icd31/SYNTHETIC_icd31.icdbackup › history.csv:127

Limitations: H17 has no calibrated thresholds and depends on motion (on a journey many cell changes are normal). Rows from before ICdetector 3.1.0 do not have it. It looks at the same network as H10 (ping-pong): agreeing with it is not an independent confirmation.

9. Data sources and quality

Declared origin: synthetic test data. Automatic signs of test data in 1 file (icd31/SYNTHETIC_icd31.icdbackup): the file name indicates synthetic data; MCC 001 (test network); operator 00101 (test network); the case device is marked as synthetic. These signs do not decide the origin.

Table 11. Files provided

File	Type	State	Content	Period
icd31/SYNTHETIC_icd31.icdbackup	Backup	Imported	9 rebuilt exports · 1,090 history rows	2026-10-15 08:00 – 2026-10-29 18:48 (phone's local time)
icd31/SYNTHETIC_icd31.icdbackup › history.csv	Observation history	Imported	1,090 rows	2026-10-15 08:00 – 2026-10-29 18:48 (phone's local time)
icd31/SYNTHETIC_icd31.icdbackup › service_state.csv	Radio: service state events	Imported	11 events	18 Oct 11:00 – 24 Oct 14:44
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	Topology	Imported	36 routes · 232 handovers (aggregate)	Routes last seen between 19 Oct 07:22 and 26 Oct 11:40 (this is not coverage)
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	Geometry	Imported	36 routes · 6 trips	Trips between 19 Oct 09:00 and 24 Oct 10:05
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	Stable-Site	Imported	3 site keys · 5 episodes	First and last aggregated record: 18 Oct 10:00 – 25 Oct 21:00
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	Forensic case	Imported	4 samples · 8 cells	28 Oct 16:12 – 28 Oct 16:13
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	Forensic case	Imported	0 samples · 0 cells	—
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	Forensic case	Imported	2 samples · 5 cells	23 Oct 13:53 – 23 Oct 13:54
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	Forensic case	Imported	5 samples · 13 cells	22 Oct 16:48 – 22 Oct 16:50

ICdetector backup

File	icd31/SYNTHETIC_icd31.icdbackup
Created by	ICdetector 3.1.0+90627b7 (database schema 26, backup format 1)
Backup time	29 Oct 19:40
Database	844.0 KiB; SHA-256 ae61065e132d132c… matches the manifest
Rows per table	history 1,090 · profile_evaluations 582 · service_state_events 11 · cell_transitions 36 · mobility_trips 6 · site_cell_evidence 11 · site_shadow_triggers 5 · forensic_cases 4 · forensic_samples 11
Rebuilt exports	Observation history; Radio: service state events; Topology; Geometry; Stable-Site; Forensic case; Forensic case; Forensic case; Forensic case
Local profile references	mature 6

CellCase rebuilds from the database the exports ICdetector 3.0.0 would have made at the time of the backup and reads them with the same readers as exported files (docs/backup-format.md). References cite those rebuilt exports: "backup › history.csv:120" is line 120 of the history exported at that time. The phone and Android version of the export are not in the backup.

Internal integrity: icd31/SYNTHETIC_icd31.icdbackup: SHA256SUMS verified (1 entry); icd31/SYNTHETIC_icd31.icdbackup › topology.zip: SHA256SUMS verified (4 entries); icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip: SHA256SUMS verified (7 entries); icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip: SHA256SUMS verified (7 entries); icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip: SHA256SUMS verified (7 entries); icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip: SHA256SUMS verified (7 entries). SHA256SUMS only detects changes made after the export; it does not prove authorship, date or chain of custody.

History: quality and time interpretation

Rows	1,090 read; 0 discarded because of their structure; 9 complete serving identities
Valid fields	GNSS 88.8% · PCI 100.0% · ARFCN 100.0% · RSRQ 100.0% · SINR 100.0% · service state 100.0%
Discarded values	0 sentinels («N/A», 2147483647…) and 0 out of range, treated as missing; the original is kept
Timing Advance	STUB_ZERO 1,090. Only LTE_INDEX and GSM_INDEX can be converted to distance; STUB_ZERO indicates a modem that constantly reports 0.
Times	1,090 with a unique UTC instant; 0 ambiguous and 0 non-existent because of a daylight saving change (not resolved)
Sampling	median between rows 15 min; 2 intervals of 60 min or more

> **Dataset cut between ICdetector 3.0 and 3.1.0**
> The history mixes 897 rows recorded before ICdetector 3.1.0 (until 26 Oct 11:50) and 193 from 3.1.0 or later (from 27 Oct 08:00), according to their AppVersion. ICdetector 3.1.0 changed how H6, H11, H12, H14, TAUnit, SecurityScore, AnomalyConfidence and the local cell confidence (and, through it, the alarms) are computed: their values are not comparable between the two parts (ICdetector CHANGELOG, «Dataset cut» marks). CellCase does not correct them and the episode priorities do not change because of it.

Rules ICdetector did not evaluate (NotEvaluatedHeuristics): 268 of 813 rows have at least one rule not evaluated for lack of data or context. H12 removed, H17 in shadow mode and H9 reserved appear in many rows, but they are not missing data and do not count as a quality problem.

Table 12. Rules not evaluated by ICdetector, by reason. Only the «missing data or context» kind is a rule that could not be evaluated.

Rule	Reason	Kind	Rows
H1 Isolated cell	no reason recorded	missing data or context	153
H6 TA distance	TA not verified yet: unknown whether the modem measures it (since 3.1.0) (H6(ta_sin_verificar))	missing data or context	39
H6 TA distance	the modem returns 0 without measuring TA (since 3.1.0) (H6(ta_stub))	missing data or context	38
H9 Ciphering (reserved)	reserved: Android does not expose ciphering (since 3.0)	reserved	515
H11 Mobile cell identity	GPS fix without known accuracy or worse than 100 m (since 3.1.0) (H11(fix_impreciso))	missing data or context	38
H11 Mobile cell identity	GPS fix older than 60 s or of unknown age (H11(fix_viejo))	missing data or context	38
H12 Latency + RF (removed since 3.1.0)	rule removed in ICdetector 3.1.0 with the latency probe; it appears in every row and is not missing data (H12(retirada))	removed	193
H17 Cell churn	shadow mode: recorded, not evaluated, no points, no thresholds until calibrated with campaign data (H17(sombra))	shadow mode	193

H12 (latency + RF) was removed in ICdetector 3.1.0 together with the latency probe: H12(retirada) appears in every 3.1.0 row and is not an abstention for lack of data.

Analyses not available

All the planned analyses have data.

Not provided: the export is missing. Not compatible: it was provided but does not have the expected structure. Import failed: the file is damaged, unreadable or was rejected for safety. Not assessable: there are data, but not enough for that analysis.

10. Methodology

Each history row with an ICdetector reason, each forensic case and each episode recorded by Stable-Site is a signal. Signals from the same cell and of the same pattern type within <= 30 min form an episode; episodes separated by <= 15 min are presented in a group, without merging them. Radio is associated with an episode if a degraded state falls within <= 5 min.

- Observation: what ICdetector recorded, quoted verbatim.
- CellCase check: comparison with the history measurements (band by EARFCN in the previous row; PCI per full cell identity and carrier within ±30 min; power against the median of the same cell outside the episode, with at least 5 rows and a difference >= 10 dB).
- Persistence (within the episode): >= 3 distinct moments over >= 10 min; signals within <= 60 s count as one moment. Recurrence: same cell and pattern on >= 2 days.
- Priority review: an ICdetector alarm (a collector warning or a forensic case opened by an alarm), complete identities, behaviour confirmed in the measurements («not contradictory» is not enough) and, in addition, persistence or recurrence.
- Review: an ICdetector alarm; or behaviour confirmed in the measurements with recurrence or persistence. An observation forensic case is not an alarm, and a coincidence with Radio is context, not independent evidence: neither raises the priority.
- Context: everything else. Verification statuses and how usual a cell is are reported, but they do not change the priority.
- Versions: readers/2.4.0, cellcase-rules/2, episodes/6, cross/3 (changes in docs/episode-rules.md and docs/cross-sources.md). Backup: backup-icd/1.1.0 (docs/backup-format.md). Local profile: profile/1, presentation only (docs/local-profile.md). Cell churn: churn/1, presentation only (docs/churn.md).

11. Conclusion and next steps

Conclusion

- Priority review: E5; review: E7, E8.
- Assessment for the spoofing question: Observations that need more data.
- No attribution to fake base stations or other attacks can be made: CellCase has no criteria implemented and evaluated for that. The lack of indications does not prove that there has been no problem at all either.

Next steps

- E5, E7: compare with captures at the exact same spot (the reference mixes different positions).
- E8: repeat the capture at the same place and time of day to see whether the pattern recurs.

12. Limitations

- ICdetector's heuristics are not calibrated; «sub-threshold» means it did not reach its alarm threshold. Its texts are quoted as written, in the original Spanish (e.g. «forzado», "forced"), and they are not CellCase conclusions.
- History, forensic case and Stable-Site come from the same collector: they are not counted as independent evidence. Radio is the only source from another family; a coincidence with Radio is reported as context, not as independent evidence, and the continuity of its log is not demonstrated.
- A single observation has an unknown duration; an interval between observations does not demonstrate continuous activity.
- Verification statuses (VERIFIED, NOT_FOUND…) are context from incomplete public databases: they neither authenticate nor rule out a base station and do not change the priority.
- The history mixes ICdetector versions: its 193 rows from 3.1.0 or later have no external verification, because ICdetector removed it, and their PENDING does not mean «pending»; only the 897 earlier rows have verification states.
- Topology, Geometry and Stable-Site are aggregates with no time per event: they describe what is usual, they do not reconstruct specific events.
- The ICdetector local profile is presented as context: shadow mode, not calibrated and not independent of the power rule (H13). It changes no priority and not the assessment.
- The device, the time zone and the origin of the data are statements by the user; no discrepancies with the files have been detected.

13. Analyst's notes

The analyst has not added notes.

Annexes

A. Files, entries and integrity

Table 13. Fingerprint of each file provided

File	File SHA-256	Size	Internal integrity
icd31/SYNTHETIC_icd31.icdbackup	`59bdeebecf83770083c4f5c7e0f5dd6f7479b601da0b1d263403e517b51097fb`	180.2 KiB	SHA256SUMS verified (1 entry)
icd31/SYNTHETIC_icd31.icdbackup › history.csv	`144a12b354167c60c0855c13fd45375ecab3d74722735144bad80dc2ee397ff7`	379.3 KiB	The format does not include SHA256SUMS
icd31/SYNTHETIC_icd31.icdbackup › service_state.csv	`6bb4b2b4d2f23254b833ae0c3372654efbfd8c84da61adac5a595ee331d63243`	1.2 KiB	The format does not include SHA256SUMS
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	`24d998957569ceb3fa387f20ed0f2af6ef23f9044b38740f4c494e3d14a19603`	14.4 KiB	SHA256SUMS verified (4 entries)
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	`37b61352433543029a8a9782aede42cef2cf1126783e05aed92c33424085dc09`	15.7 KiB	The format does not include SHA256SUMS
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	`62af6434e9068437c86c1222eca6cb3b875524e954e56a90aa8fdc866f101c8c`	4.6 KiB	The format does not include SHA256SUMS
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	`3252955579cb524a0e284bd6440a571e39ccd15ad1f5b5c0e286850463437b8c`	6.9 KiB	SHA256SUMS verified (7 entries)
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	`a908302f9c5d198f636c4a69cdd49037fdaf4b6a0e00a02b15030d1ed47da7e0`	1.8 KiB	SHA256SUMS verified (7 entries)
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	`4d8421f6c05c42978b146b51f60828de66420bd4a0f2712cd9aaccd341f7fe82`	7.9 KiB	SHA256SUMS verified (7 entries)
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	`0a469a20b35b1ab535c6d4b0738111fde8947b4b26cb9e47aa2781d391e0fbd6`	13.8 KiB	SHA256SUMS verified (7 entries)

SHA-256 computed by CellCase over the bytes of the file when it was imported. It allows checking that the stored copy has not changed; it does not prove authenticity, authorship, capture date or chain of custody.

Table 14. Internal ZIP entries

File	Entry	Role	Rows	Size
icd31/SYNTHETIC_icd31.icdbackup	manifest.json	Metadata	—	275 B
icd31/SYNTHETIC_icd31.icdbackup	icdetector_history.db	Data	—	844.0 KiB
icd31/SYNTHETIC_icd31.icdbackup	settings.json	Context	—	169 B
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	cells.csv	Check	9	587 B
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	transitions.csv	Data	36	3.8 KiB

Table 14. Internal ZIP entries (continued)

File	Entry	Role	Rows	Size
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	topology.graphml	Check	—	9.3 KiB
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	metadata.json	Metadata	—	415 B
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	SHA256SUMS.txt	Integrity	—	321 B
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	geometry_cells.csv	Data	9	1.2 KiB
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	geometry_edges.csv	Data	36	3.6 KiB
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	mobility_trips.csv	Data	6	696 B
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	geometry.graphml	Check	—	9.4 KiB
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	metadata.json	Metadata	—	860 B
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	PRIVACY.txt	Context	—	905 B
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	sites.csv	Data	3	619 B
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	site_cells.csv	Data	6	817 B
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	site_rf_neighbours.csv	Context	2	343 B
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	motion.csv	Data	15	927 B
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	shadow_episodes.csv	Data	5	1.1 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	case.json	Metadata	—	569 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	timeline.csv	Data	4	1.3 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	cells.csv	Data	8	1.6 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	heuristics.csv	Data	8	731 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	trust_site.csv	Context	4	926 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	capabilities.json	Context	—	992 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	terminal.log	Context	—	340 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	SHA256SUMS.txt	Integrity	—	556 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	case.json	Metadata	—	551 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	timeline.csv	Data	0	184 B

Table 14. Internal ZIP entries (continued)

File	Entry	Role	Rows	Size
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	cells.csv	Data	0	184 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	heuristics.csv	Data	0	51 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	trust_site.csv	Context	0	262 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	capabilities.json	Context	—	46 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	terminal.log	Context	—	0 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	SHA256SUMS.txt	Integrity	—	556 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	case.json	Metadata	—	593 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	timeline.csv	Data	2	691 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	cells.csv	Data	5	1.1 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	heuristics.csv	Data	4	391 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	trust_site.csv	Context	2	594 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	capabilities.json	Context	—	520 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	terminal.log	Context	—	3.5 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	SHA256SUMS.txt	Integrity	—	556 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	case.json	Metadata	—	568 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	timeline.csv	Data	5	1.4 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	cells.csv	Data	13	2.6 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	heuristics.csv	Data	10	941 B
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	trust_site.csv	Context	5	1.0 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	capabilities.json	Context	—	1.2 KiB

Table 14. Internal ZIP entries (continued)

File	Entry	Role	Rows	Size
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	terminal.log	Context	—	5.5 KiB
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	SHA256SUMS.txt	Integrity	—	556 B

B. Validation issues

Table 15. Issues grouped by file

File	Code	Severity	Rows	Description and examples
icd31/SYNTHETIC_icd31.icdbackup	BACKUP_RECONSTRUCTED	Info	1	Exports rebuilt from the database with the ICdetector 3.0.0 rules, at the time of the backup (2026-10-29T18:40:11.500Z). ExportDevice and ExportAndroid are not in the backup and are left empty.
icd31/SYNTHETIC_icd31.icdbackup › history.csv	DUP_SAME_OBSERVATION	Info	1	Same reading (ObservationId) as another row: ICdetector sometimes saves the periodic row and the confirmed-alarm row of the same reading. Neither is removed. E.g.: line 494 «b113a7ed-f3f3-4ddf-ac2b-ce30ab9ec132» (see line 493).
icd31/SYNTHETIC_icd31.icdbackup › history.csv	TEXT_GUARD_REMOVED	Info	3	Text with the ICdetector 3.0 protective apostrophe (it stops a spreadsheet from reading it as a formula): it is read without it. The example shows the value as it is in the file. E.g.: line 359 «"ya protegido»; line 361 «'-Femto»; line 362 «'=Femto,"casa"».
icd31/SYNTHETIC_icd31.icdbackup › history.csv	TIME_AMBIGUOUS_RESOLVED	Info	9	Local time repeated by the clock change, resolved with ObservedAtUtc. E.g.: line 304 «2026-10-25 02:48:27»; line 305 «2026-10-25 02:30:00»; line 306 «2026-10-25 02:17:56».
icd31/SYNTHETIC_icd31.icdbackup › service_state.csv	TEXT_GUARD_REMOVED	Info	5	Text with the ICdetector 3.0 protective apostrophe (it stops a spreadsheet from reading it as a formula): it is read without it. The example shows the value as it is in the file. E.g.: line 4 «'-Red de Pruebas-»; line 6 «'-Red de Pruebas-»; line 8 «'-Red de Pruebas-».
icd31/SYNTHETIC_icd31.icdbackup › topology.zip	SCOPE_AGGREGATE	Info	1	Cumulative aggregate: totals per route and last time seen. It contains neither individual handovers nor the period the counters cover.
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	SCOPE_AGGREGATE	Info	1	Aggregate of routes and trips without coordinates; trip_count may include trips already pruned (older than 90 days).
icd31/SYNTHETIC_icd31.icdbackup › geometry.zip	TRIPS_DISCARDED	Info	1	2 CLOSED trips do not meet the validity rule (movement and 3 cells): ICdetector discards them but still exports them as CLOSED.
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: evidence_type, radio, arfcn, pci, mcc_metadata, mnc_metadata, tac_metadata, observations, distinct_days, first_seen_ms, last_seen_ms.
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip	SCOPE_OVERLAP	Info	1	Each place is recorded under up to 4 overlapping site keys: per-site counters are not added up as unique observations.
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: connection_state, bandwidth_khz.

Table 15. Issues grouped by file (continued)

File	Code	Severity	Rows	Description and examples
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: site_novelty, site_enforced, site_static_duration_s, site_location_accuracy_m, site_serving_days, site_neighbour_days.
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	CASE_INCOMPLETE	Warning	1	Case exported in state CAPTURING: the capture had not finished (D-013).
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: connection_state, bandwidth_khz.
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-26-0003.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: site_novelty, site_enforced, site_static_duration_s, site_location_accuracy_m, site_serving_days, site_neighbour_days.
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: connection_state, bandwidth_khz.
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: site_novelty, site_enforced, site_static_duration_s, site_location_accuracy_m, site_serving_days, site_neighbour_days.
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip	TEXT_GUARD_REMOVED	Info	2	Text with the ICdetector 3.0 protective apostrophe (it stops a spreadsheet from reading it as a formula): it is read without it. The example shows the value as it is in the file. E.g.: line 2 «'+19 dB sobre la referencia, "sin" conte»; line 4 «'+19 dB sobre la referencia, "sin" conte».
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: connection_state, bandwidth_khz.
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: site_novelty, site_enforced, site_static_duration_s, site_location_accuracy_m, site_serving_days, site_neighbour_days.
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip	TEXT_GUARD_REMOVED	Info	5	Text with the ICdetector 3.0 protective apostrophe (it stops a spreadsheet from reading it as a formula): it is read without it. The example shows the value as it is in the file. E.g.: line 2 «'+19 dB sobre la referencia, "sin" conte»; line 4 «'+19 dB sobre la referencia, "sin" conte»; line 6 «'+19 dB sobre la referencia, "sin" conte».

C. Evidence references by episode

Each row is identified by file and physical line (the header is line 1) so it can be found in the original. Coordinates with visual rounding (annex F).

Table 16. E1: rows with a signal and the row before each one; the rest of the interval (5 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › history.csv:749	2026-10-19 07:22:55	`001-01-3001-1000980-LTE`	-98	OK	—
icd31/SYNTHETIC_icd31.icdbackup › history.csv:748	2026-10-19 07:23:20	`001-01-3001-1000069-LTE`	-91	[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)	0.009, 0.022

Table 17. E2: rows with a signal and the row before each one; the rest of the interval (3 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip!shadow_episodes.csv:4	2026-10-21 03:11:54	`001-01-3001-1000070-LTE`	—	Stable-Site: NOVEL_CELL	—

Table 18. E3: rows with a signal and the row before each one; the rest of the interval (7 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › history.csv:587	2026-10-21 12:01:47	`001-01-3002-1000544-LTE`	-90	OK	0.012, 0.020
icd31/SYNTHETIC_icd31.icdbackup › history.csv:586	2026-10-21 12:02:12	`001-01-3002-1000545-LTE`	-91	[sub-umbral] Downgrade de banda forzado (2620MHz->925MHz, B7->B8)	—
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip!shadow_episodes.csv:5	2026-10-21 12:02:12	`001-01-3002-1000545-LTE`	—	Stable-Site: KNOWN_CELL	—

Table 19. E4: rows with a signal and the row before each one; the rest of the interval (6 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › history.csv:583	2026-10-21 12:32:42	`001-01-3002-1000542-LTE`	-85	OK	0.009, 0.021
icd31/SYNTHETIC_icd31.icdbackup › history.csv:582	2026-10-21 12:33:07	`001-01-3002-1000543-LTE`	-83	[sub-umbral] Downgrade de banda forzado (2620MHz->791MHz, B7->B20)	0.011, 0.021
icd31/SYNTHETIC_icd31.icdbackup › stable_site.zip!shadow_episodes.csv:6	2026-10-21 12:33:07	`001-01-3002-1000543-LTE`	—	Stable-Site: NOVEL_CELL	—

Table 20. E5: rows with a signal and the row before each one; the rest of the interval (6 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › history.csv:495	2026-10-22 16:46:47	`001-01-3001-1000980-LTE`	-96	OK	—
icd31/SYNTHETIC_icd31.icdbackup › history.csv:493	2026-10-22 16:48:53	`001-01-3001-1000070-LTE`	-81	[sub-umbral] Potencia anómala vs historial (+19dB) [confirmada 3/3]	0.010, 0.018
icd31/SYNTHETIC_icd31.icdbackup › history.csv:494	2026-10-22 16:48:53	`001-01-3001-1000070-LTE`	-81	[sub-umbral] Potencia anómala vs historial (+19dB)	0.010, 0.018
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip!timeline.csv:2; icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-22-0001.zip!timeline.csv:6	2026-10-22 16:48:53	`001-01-3001-1000070-LTE`	—	forensic case: ICD-2026-10-22-0001	—

Table 21. E6: rows with a signal and the row before each one; the rest of the interval (7 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › history.csv:430	2026-10-23 13:53:20	`001-01-3003-1000953-LTE`	-103	OK	0.030, 0.040
icd31/SYNTHETIC_icd31.icdbackup › history.csv:429	2026-10-23 13:53:44	`001-01-3003-1000953-LTE`	-109	[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)	0.029, 0.040
icd31/SYNTHETIC_icd31.icdbackup › history.csv:428	2026-10-23 13:54:10	`001-01-3003-1000953-LTE`	-113	OK	0.029, 0.039
icd31/SYNTHETIC_icd31.icdbackup › history.csv:427	2026-10-23 13:54:36	`001-01-3003-1000953-LTE`	-102	[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)	0.029, 0.039
icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip!timeline.csv:2; icd31/SYNTHETIC_icd31.icdbackup › ICD-OBS-2026-10-23-0002.zip!timeline.csv:3	2026-10-23 13:53:44	`001-01-3003-1000953-LTE`	—	forensic case: ICD-OBS-2026-10-23-0002	—

Table 22. E7: rows with a signal and the row before each one; the rest of the interval (4 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › history.csv:170	2026-10-27 15:15:52	001-01-3001-1000980-LTE	-97	OK	0.012, 0.019
icd31/SYNTHETIC_icd31.icdbackup › history.csv:169	2026-10-27 15:31:20	001-01-3001-1000070-LTE	-79	[sub-umbral] Potencia anómala vs historial (+21dB)	0.011, 0.020

Table 23. E8: rows with a signal and the row before each one; the rest of the interval (3 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip!timeline.csv:2 ; icd31/SYNTHETIC_icd31.icdbackup › ICD-2026-10-28-0004.zip!timeline.csv:5	2026-10-28 16:12:00	001-01-3002-1000542-LTE	—	forensic case: ICD-2026-10-28-0004	—

D. Statistical description of the history

Table 24. Serving identities with the most rows (9 of 9)

Identity	Rows	% rows	First (local)	Last (local)
001-01-3001-1000070-LTE	712	65.3%	2026-10-15 08:10:05	2026-10-29 18:18:13
001-01-3001-1000980-LTE	106	9.7%	2026-10-15 11:54:15	2026-10-29 15:40:41
001-01-3001-1000069-LTE	87	8.0%	2026-10-15 08:00:03	2026-10-29 17:47:32
001-01-3001-1000071-LTE	87	8.0%	2026-10-15 11:43:35	2026-10-29 18:48:53
001-01-3002-1000544-LTE	48	4.4%	2026-10-18 11:55:48	2026-10-29 14:13:43
001-01-3002-1000542-LTE	44	4.0%	2026-10-18 18:53:42	2026-10-29 14:54:47
001-01-3003-1000953-LTE	4	0.4%	2026-10-23 13:53:20	2026-10-23 13:54:36
001-01-3002-1000543-LTE	1	0.1%	2026-10-21 12:33:07	2026-10-21 12:33:07
001-01-3002-1000545-LTE	1	0.1%	2026-10-21 12:02:12	2026-10-21 12:02:12

Table 25. Recorded power (dBm) by technology; technologies are not combined

Technology	Quantity	n	Min.	P10	Median	P90	Max.
LTE	getDbm(), RSRP on current Android	1,090	-113	-104	-98	-91	-79

ICdetector results	OK 1,082 · sub-threshold 8 · reason without prefix 0 · SecurityScore < 100 in 8 rows
AnomalyConfidence	maximum 10.2: an uncalibrated internal score, not a probability
External verification	NOT_FOUND 678; PENDING 32; VERIFIED 187; no external verification (removed in ICdetector 3.1.0) 193
TA consistency	0 rows with TAMeters different from the documented conversion
Duplicates in the file	0 identical; 2 with the same time and identity; none removed
Interval with no records	threshold 60 min

E. Rules and thresholds

- Grouping: same cell and same pattern type within <= 30 min; groups of episodes within <= 15 min (without merging them); Radio within <= 5 min.
- Observation moment: signals within <= 60 s are counted once. Persistence: >= 3 moments over >= 10 min. Recurrence: same cell and pattern on >= 2 days.
- LTE band: Bands column or EARFCN (3GPP TS 36.101); compared with the previous row within <= 10 min.
- Power: median of the same cell outside ±30 min of the episode, with >= 5 rows; significant difference >= 10 dB.
- Evidence families: collector (history, forensic case, Stable-Site) and phone state (Radio). Radio is the only separate family; a coincidence with Radio is reported as context and does not raise the priority.
- Interval with no records: 60 min or more between rows with an exact time. Valid ranges and sentinels: docs/analysis-rules.md.
- Versions: cellcase-rules/2, episodes/6, readers/2.4.0, cross/3.

F. Privacy and rounding

- Coordinates: rounded to 3 decimals (around 100 m).
- Cell identifiers: complete.
- Powers and percentages are shown with the rounding stated in each table; the calculations use the original values.
- The original files kept in the app are not modified.